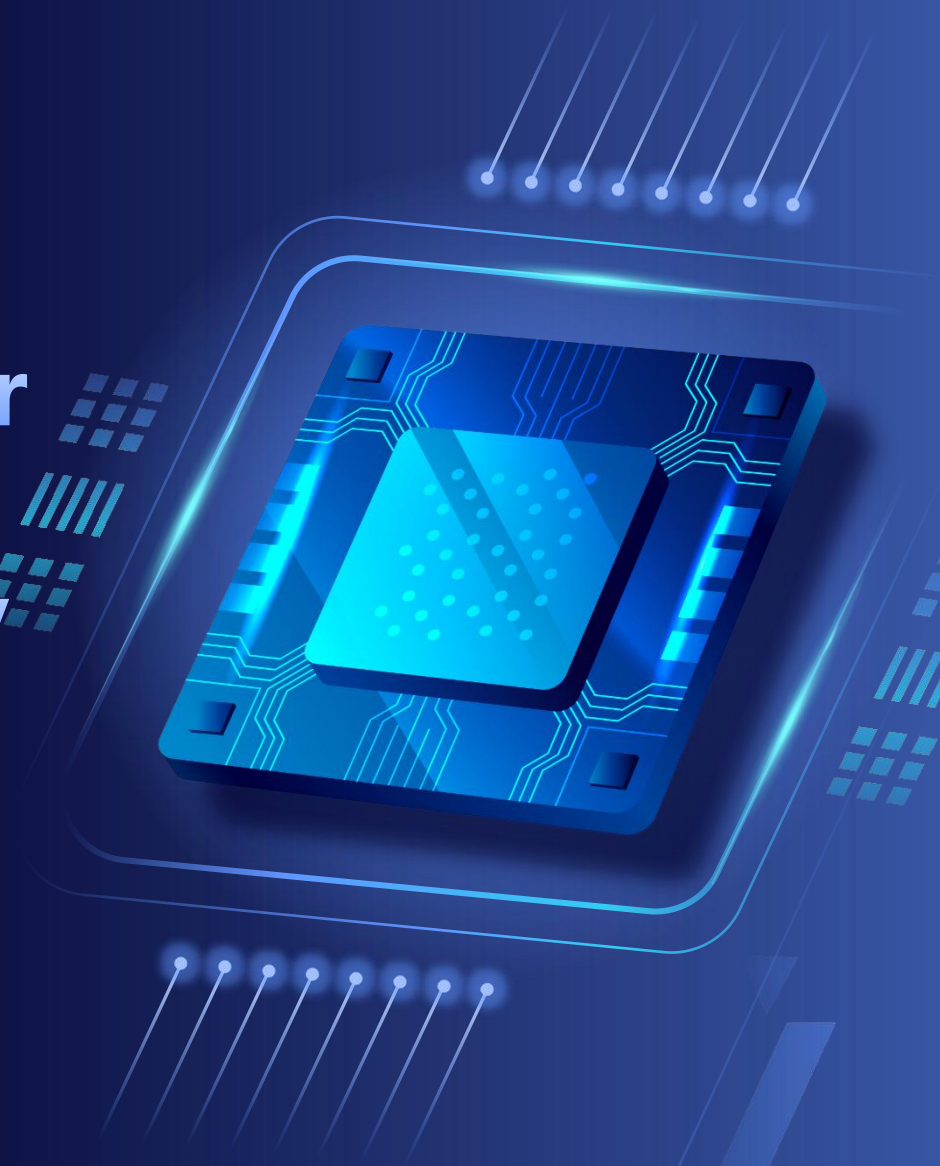
High-Power GaN Blue Laser Diodes: Degradation Mechanisms and Reliability Enhancement



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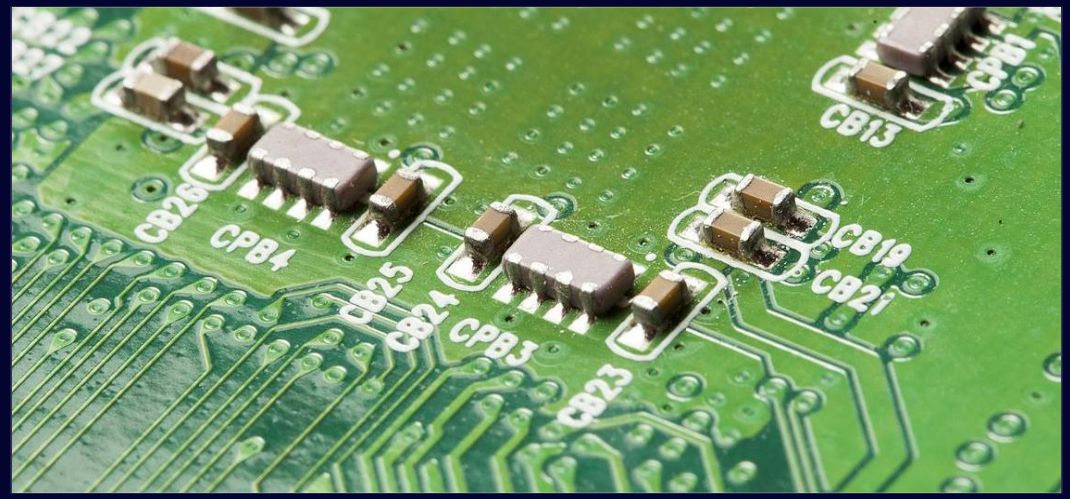
01

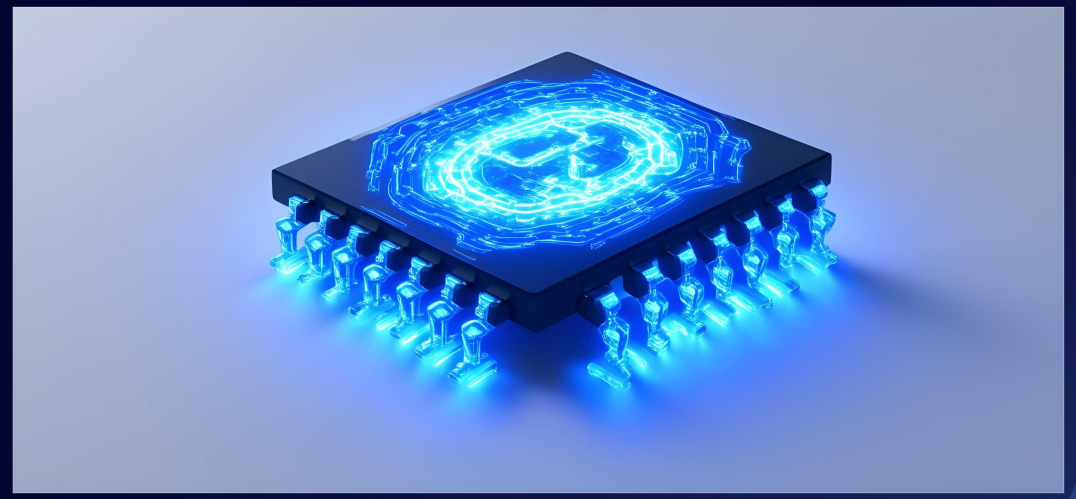
Fundamentals of GaN Blue LDs

Working Principles

Electroluminescence in InGaN/GaN MQWs

When a forward bias is applied to the GaN - based laser diode, electrons and holes are injected into the InGaN/GaN multiple quantum well (MQW) structure. In the MQWs, electrons and holes recombine radiatively, emitting photons. The bandgap of InGaN can be adjusted by changing the indium composition, enabling the emission of blue light. For example, in a typical 437 nm **GaN - based blue LD**, the MQW structure is carefully designed to achieve efficient radiative recombination.

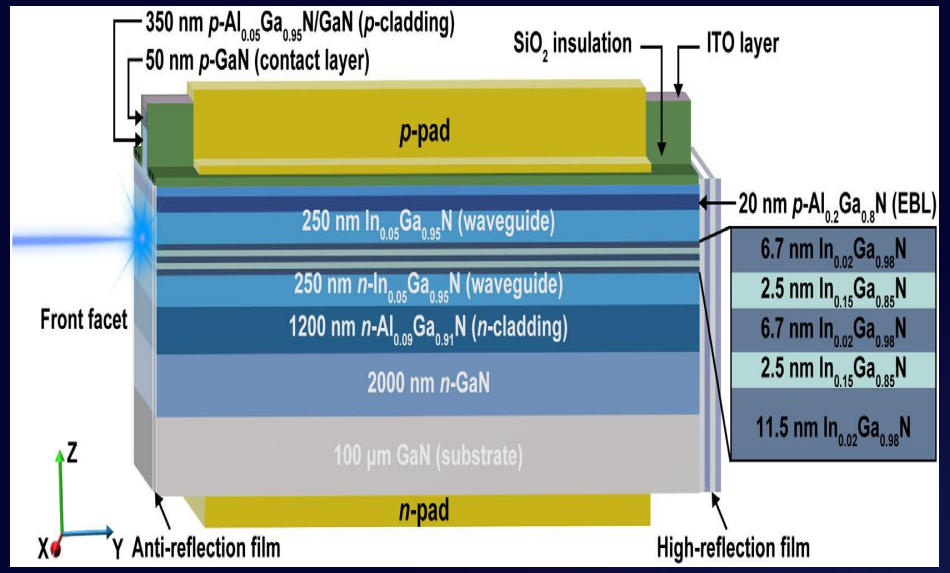




Material Systems

GaN Substrate Epitaxy

GaN substrates play a crucial role in the performance of blue laser diodes. Epitaxial growth on GaN substrates, such as free - standing GaN substrates with a dislocation density less than (3times10^{6} cm^{-2}), provides a high - quality foundation for the subsequent layers. The crystallographic orientation of the substrate affects the growth of the epitaxial layers. For instance, c - plane GaN substrates are commonly used, and the epitaxial growth process can be precisely controlled to minimize defects and improve device performance.



The structure diagram of the ridge-waveguide edge-emitting LD.

Applications

Display Applications

In the display industry, GaN - based blue LDs are essential for laser - based display technologies. They can provide high - brightness and high - color - gamut light sources. For instance, in some high - end laser projectors, GaN blue LDs with output powers around 3 - 5 W contribute to vivid and sharp images, enhancing the visual experience of users.

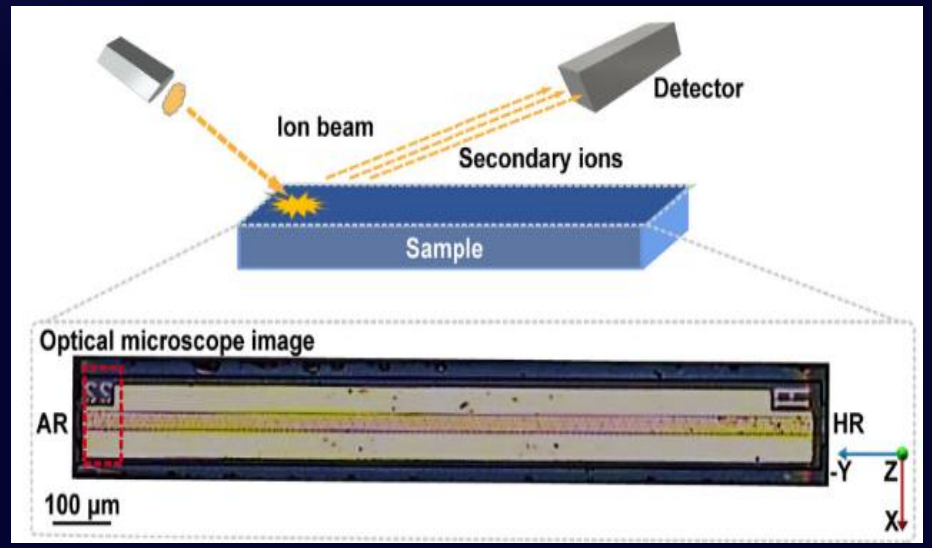
Industrial Applications

High - power GaN - based blue laser diodes are widely used in industrial applications. In laser cutting and welding, their high power density enables precise processing of various materials. For example, in the manufacturing of electronic components, 5 - 7.5 W GaN blue LDs can cut thin metal sheets with high precision, improving production efficiency and product quality.

02

Degradation Mechanisms

Optical Loss & Impurity Diffusion



The schematic diagram of SIMS analysis.

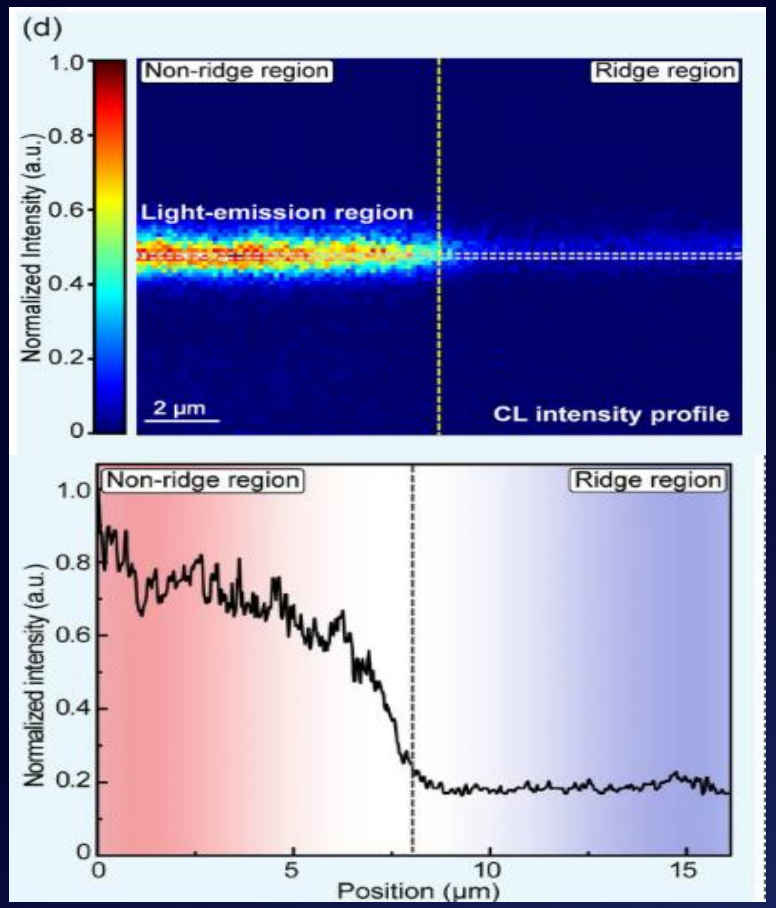
Analysis of SiO_2 Diffusion by SIMS

Secondary Ion Mass Spectroscopy (SIMS) analysis revealed significant Si diffusion in the p - type layer of the LD. The diffusion depth of Si from the SiO_2 mirror coating was nearly 500 nm, suggesting that the diffusion of SiO_2 mirror coating from the cavity surface into the p - type layer is a key factor contributing to optical loss and LD degradation.

Optical Loss & Impurity Diffusion

Quantification of Point Defects by CL

Using Cathodoluminescence (CL), a significant decrease in CL intensity was observed in the ridge region of the LD. For example, the CL intensity in the ridge region dropped to only about 26% of that in the non - ridge region, indicating an increase in point defects acting as non - radiative recombination centers (NRCs) and a reduction in internal quantum efficiency (IQE).



Normalized intensity of CL signal

Thermal-Electrical Coupling

The PICS3D commercial software was used to simulate the L - I - V characteristics of the LD. In the simulations.

Modeling Threshold Current Rise

Through PICS3D simulations, it was found that as the temperature increased, the threshold current of the LD rose. For instance, with the increase of junction temperature, the threshold current increased, which was consistent with the experimental results, indicating that temperature has a significant impact on the electrical characteristics of the LD.

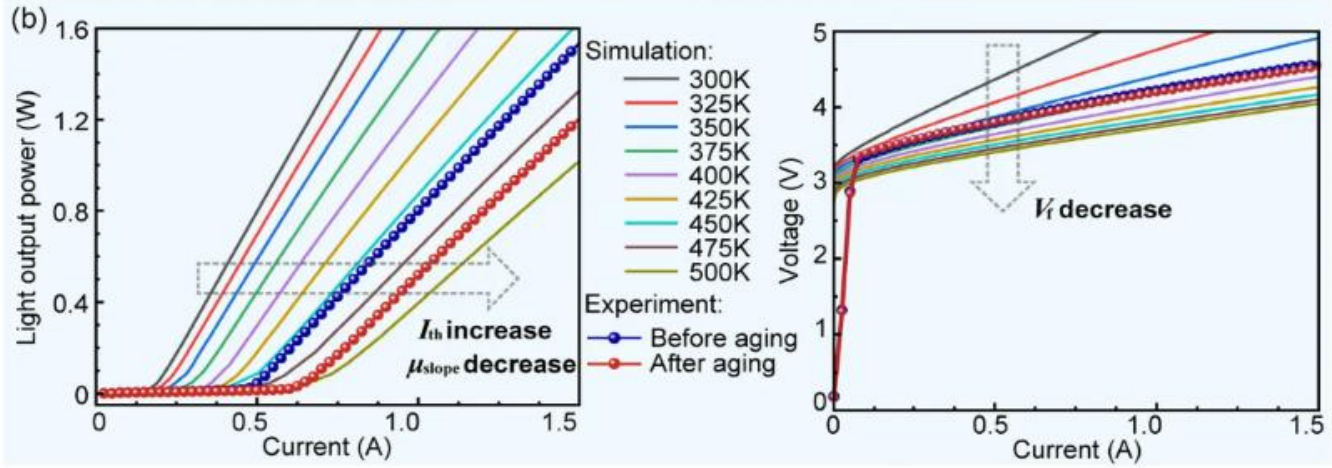


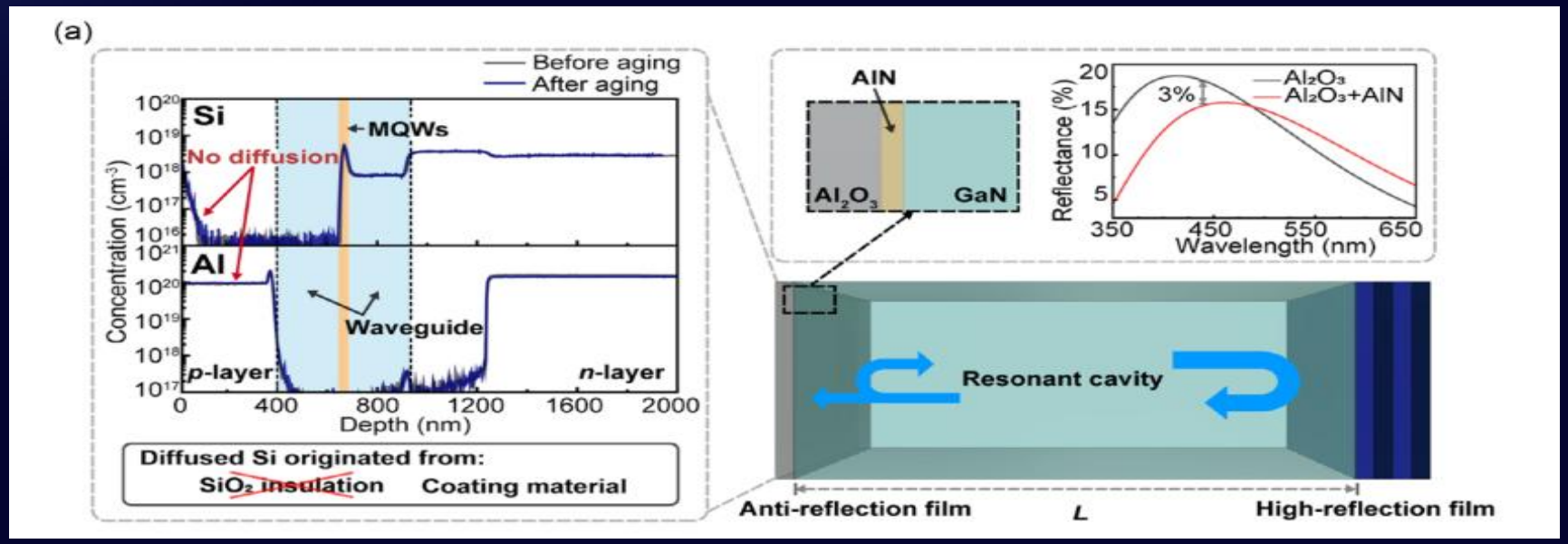
Figure 5. a) Simulated data of I–V characteristics with and without Si diffusion. b) The simulated and experimental data of L–I and I–V characteristics.

Temperature coefficients play a crucial role in the simulation. They help to elucidate how temperature affects the thermal escape of injected carriers, non - radiative recombination, optical loss, and quantum efficiency, ultimately leading to changes in the L - I - V characteristics of the LD.

03

Anti-Aging Solutions

AlN/Al_2O_3 Interface Engineering



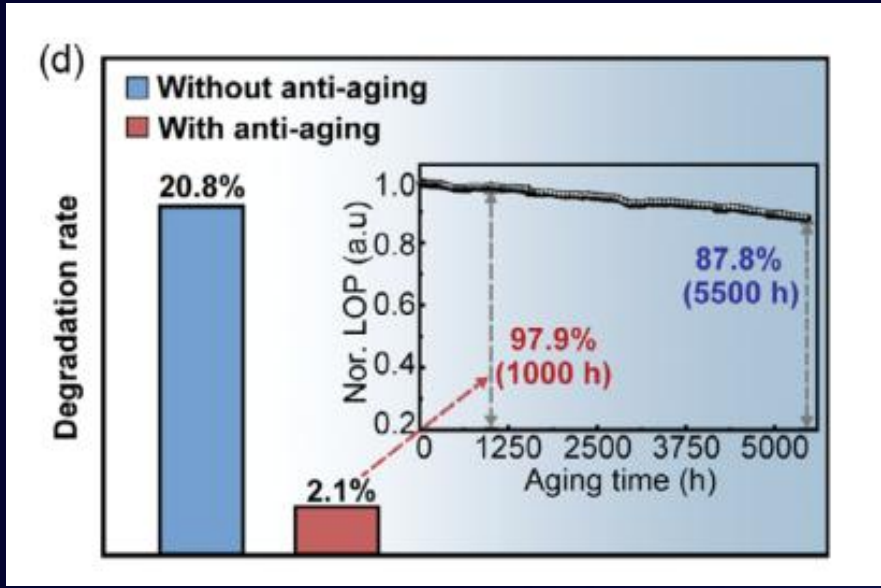
The Role of AlN as a Diffusion - blocking Layer

AlN is introduced between GaN and the mirror film. Its wide band - gap of ≈6.1 eV effectively prevents electron - hole recombination on the cavity surface. High melting point (2473 K) and thermal conductivity (170 W m^{-1} K^{-1}) act as a barrier to suppress the diffusion of SiO_2 mirror coating from the cavity surface into the LD epilayer, as confirmed by SIMS analysis showing no Al diffusion from Al_2O_3 mirror film into the p - type layer.

AlN/Al_2O_3 Interface Engineering

Enhanced Reliability with Al_2O_3 Mirror Film

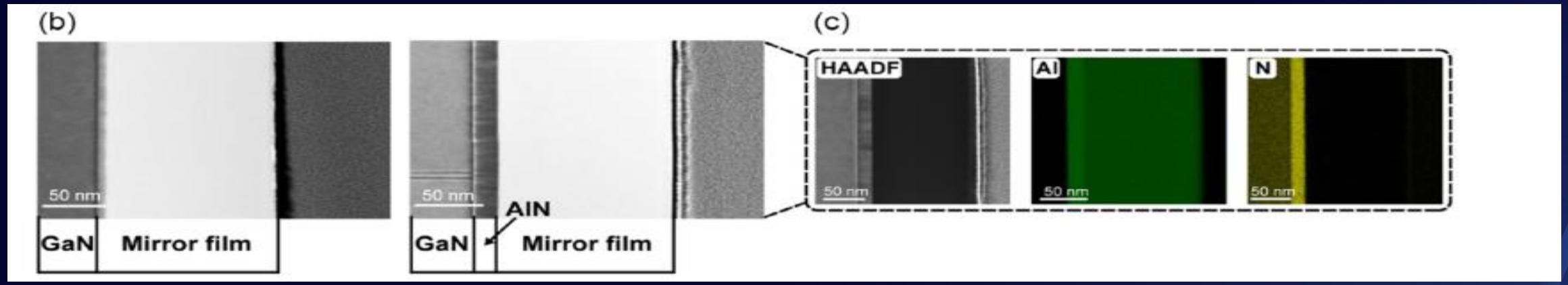
Al_2O_3 is used to replace SiO_2 as the mirror film material. With a melting point of 2323 K (673 K higher than SiO_2) and thermal conductivity 5.9 times that of SiO_2, it can better withstand the high - power density and increased temperature near the cavity surface, improving the reliability of the mirror film.



AlN/Al_2O_3 Interface Engineering

TEM Evidence of Degradation Suppression

TEM images of the GaN - mirror film interface with the AlN layer show a clear boundary. Aging test results indicate that the sample with this AlN/Al_2O_3 structure has nearly 10 - times degradation suppression compared to samples without this technology. For example, the normalized light output power of the sample with this technology decreased to 97.9% at 1000 h of aging, showing a significant improvement in lifespan.



Hybrid ITO Cladding

Refractive Index Optimization for Internal Loss Reduction

The refractive index of ITO is much lower than that of the p - AlGaN cladding layer. By using ITO to replace part of the conventional p - AlGaN cladding layer, it provides more sufficient optical confinement. Simulations show that the internal loss of hybrid ITO LDs can be reduced. Our internal loss for hybrid LDs is almost one order of magnitude lower than that reported by some other groups, reaching as low as 1.4 cm^{-1}.

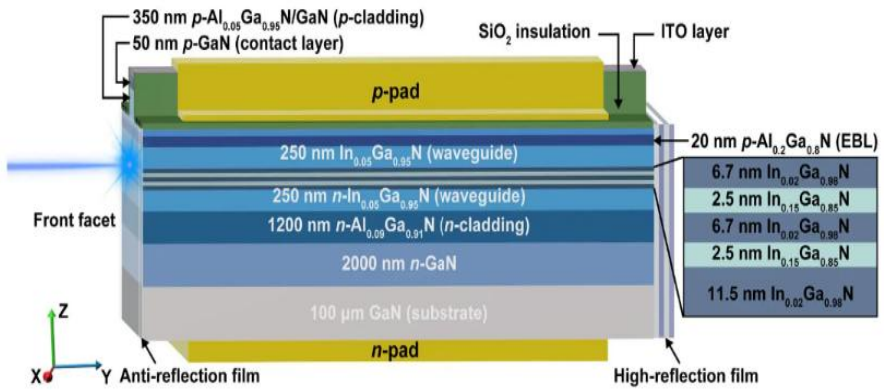


Figure 1. The structure diagram of the ridge-waveguide edge-emitting LD. The ridge-waveguide structure was 30 × 1200 μm^2. The *p*-pad and *n*-pad were deposited on each side for the current injection. To form the *p*-type contact, an indium tin oxide (ITO) layer was deposited. The electron-blocking layer (EBL) was grown between the *p*-cladding layer and the waveguide layer.

Hybrid ITO Cladding

Advantages over Conventional Cladding Structures

Compared to conventional p - AlGaN cladding layers, ITO cladding has several advantages. It has much lower resistance, which reduces power consumption. The deposition temperature of ITO can be around 300 °C or even room temperature, reducing the thermal budget imposed on InGaN QWs and thus minimizing thermal degradation of InGaN QWs. Also, its absorption coefficient is 2 orders of magnitude lower than that of metal, further contributing to the reduction of internal loss.

Achieving High Wall - Plug Efficiency (WPE)
With the reduction of internal loss, the hybrid GaN - based LDs with ITO cladding can achieve a high wall - plug efficiency. At an operation current of 3 A, an output power of 5 W and a wall - plug efficiency of 41% have been achieved.

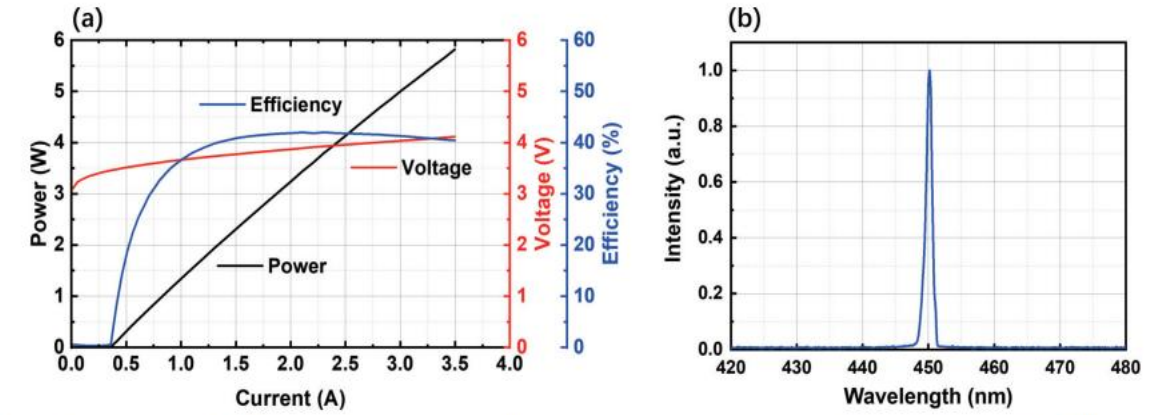


Fig. 3. (Color online) (a) *P–I–V* curves, efficiency curve, and (b) lasing spectra of optimized hybrid GaN-based LDs with ITO cladding layer.

04

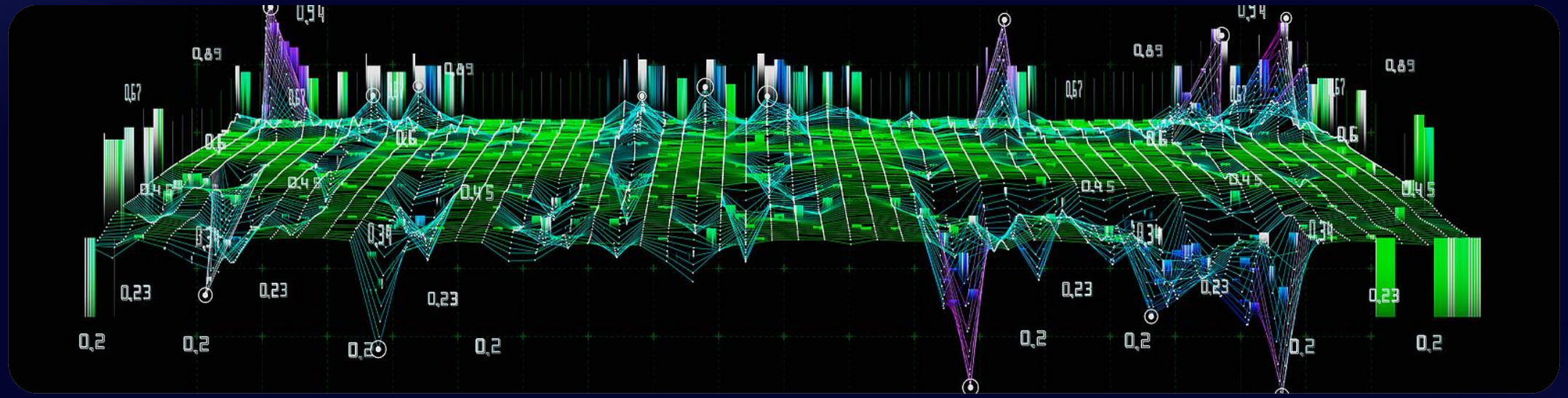
Performance Enhancement

Epitaxial & Thermal Design

Optimizing Carrier Injection with 12nm EBL

The 12nm-thin AlGaN electron - blocking layer (EBL) plays a crucial role. It effectively balances electron injection and optical confinement. By precisely tuning the thickness, it ensures that electrons are efficiently blocked from overflowing, enhancing the radiative recombination in the active region, thus improving the overall performance of the laser diode.

Epitaxial & Thermal Design



Microchannel Coolers for Thermal Resistance Reduction

Microchannel coolers are key components in thermal management. They are designed to achieve a thermal resistance as low as 8 $K \cdot W^{-1}$. These coolers work by rapidly dissipating heat generated during the operation of the high - power GaN - based blue laser diodes, maintaining a stable operating temperature and enhancing device reliability.

Efficiency Breakthroughs

Asymmetric MQW for Efficiency Improvement

Asymmetric multiple quantum well (MQW) structures are utilized in the prototypes achieving 55% wall - plug efficiency (WPE). This design allows for better control of carrier distribution and optical gain, optimizing the conversion of electrical energy into optical energy, thereby significantly increasing the efficiency of the laser diode.

05

Experimental Validation

Aging Test Protocol

The 5,500 - hour aging test at 60°C is significant as it helps in understanding the long - term stability and degradation mechanisms of the GaN - based blue laser diodes. It serves as a key step in validating the effectiveness of anti - aging technologies and predicting the device's lifespan under practical conditions.

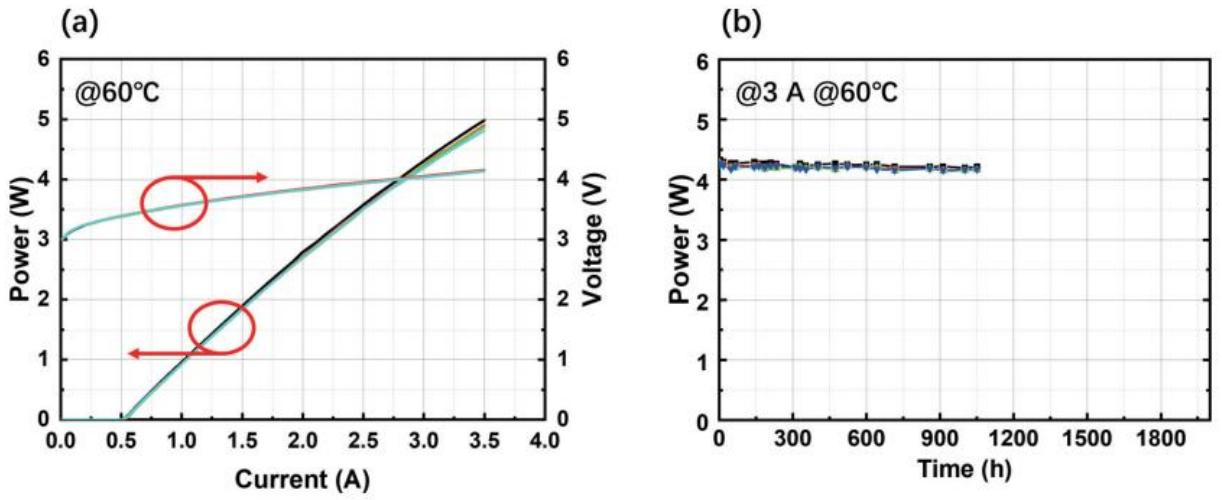


Fig. 4. (Color online) (a) *P–I–V* curves and (b) aging curves of optimized hybrid GaN-based LDs with ITO cladding layer at 60 °C case temperature.

Reliability Verification

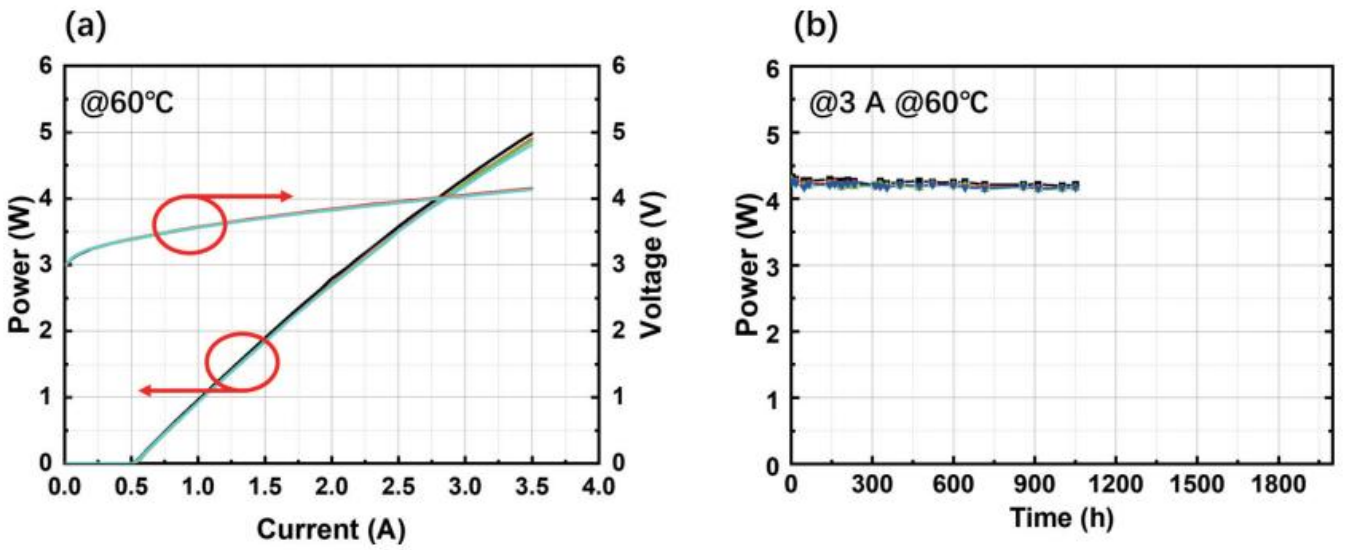


Fig. 4. (Color online) (a) *P–I–V* curves and (b) aging curves of optimized hybrid GaN-based LDs with ITO cladding layer at 60 °C case temperature.

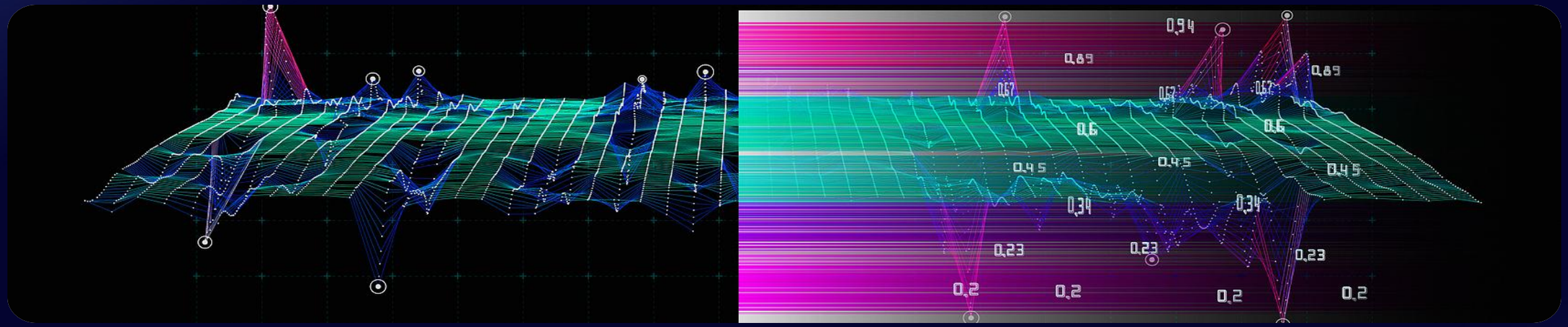
Implications for Device Reliability

The confirmation of a >20,000 - hour lifetime at 5W output power under industrial - like conditions has significant implications for the reliability of the devices. It indicates that the anti - aging technologies and design optimizations implemented are effective in enhancing the long - term performance and reliability of high - power GaN - based blue laser diodes.

06

Conclusion & Prospects

Key Findings

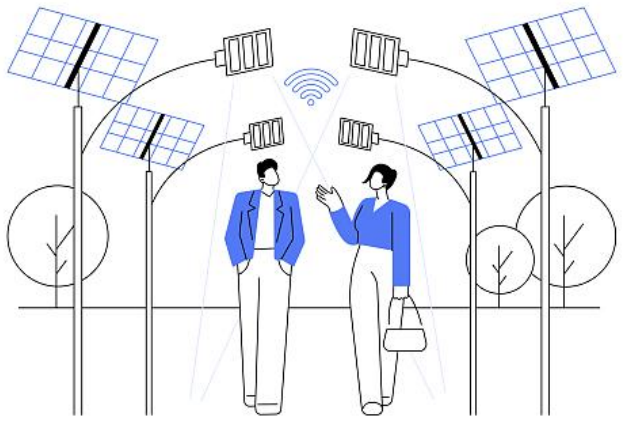


Significant progress has been made in understanding and mitigating the degradation mechanisms of high-power GaN-based blue laser diodes. The SiO_2 diffusion into the p-type layer has been identified as a key degradation factor, and the development of anti-aging technologies, such as the AlN passivation layer and Al_2O_3 mirror films, has shown significant improvements in lifespan and reliability.

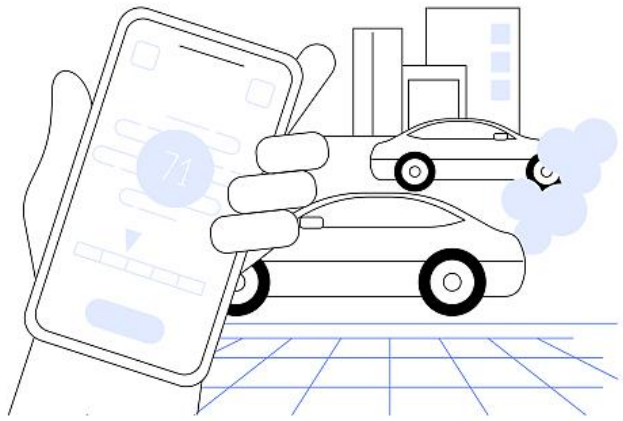
Future Roadmap

20W Pulsed Operation Goal

Looking ahead, one of the key future directions is to achieve 20W pulsed operation. This would open up new applications in areas such as high - energy laser processing and advanced optical communication systems that require high - peak - power light sources.



SMART CITY STREET LIGHTING



AIR QUALITY MONITORING

TRAFFIC DATA

Future Roadmap

Wall - Plug Efficiency Target

Another important goal is to increase the wall - plug efficiency (WPE) to over 60%. Improving WPE not only enhances the energy - conversion efficiency of the LDs but also reduces power consumption and heat generation, which is crucial for their long - term reliable operation.

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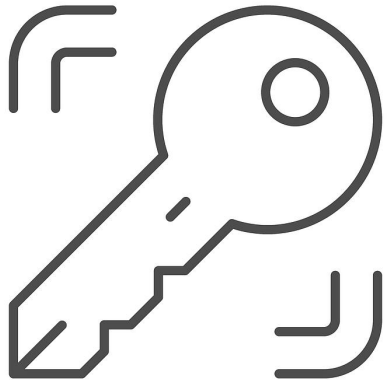

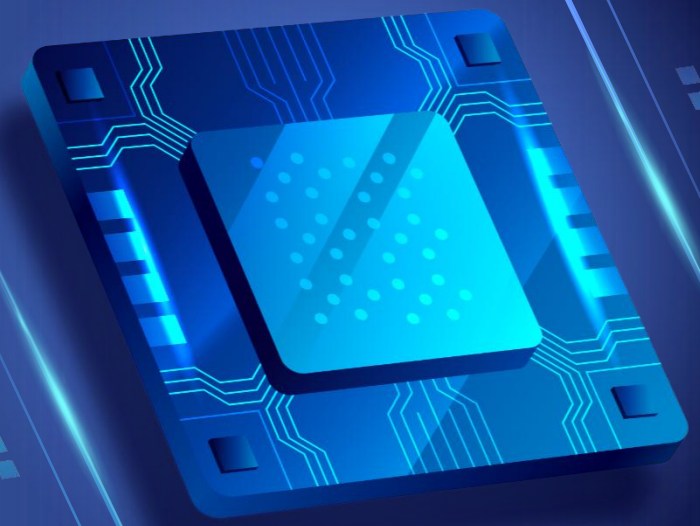

THE END

THANKS